

ASSIGNMENT ON CSE-260

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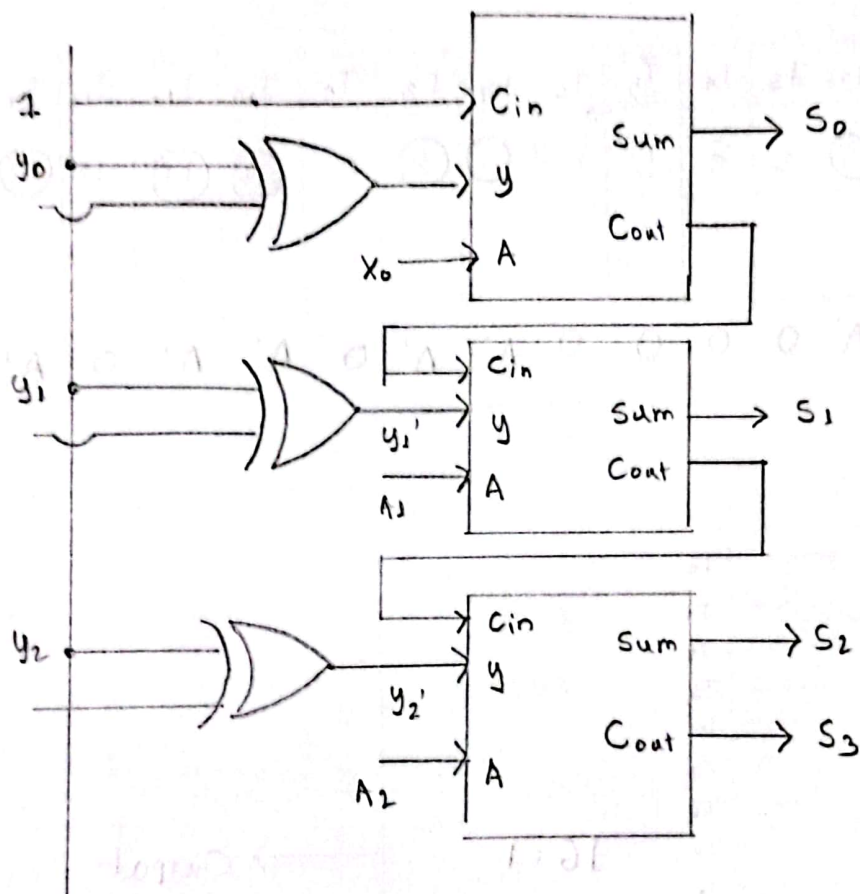
Section : 10

Assignment : 3

Date : 21 / 1 / 25

Answer to the Question No. 1

So, a circuit implementing L 's complement using encoder and decoder has been built below:-



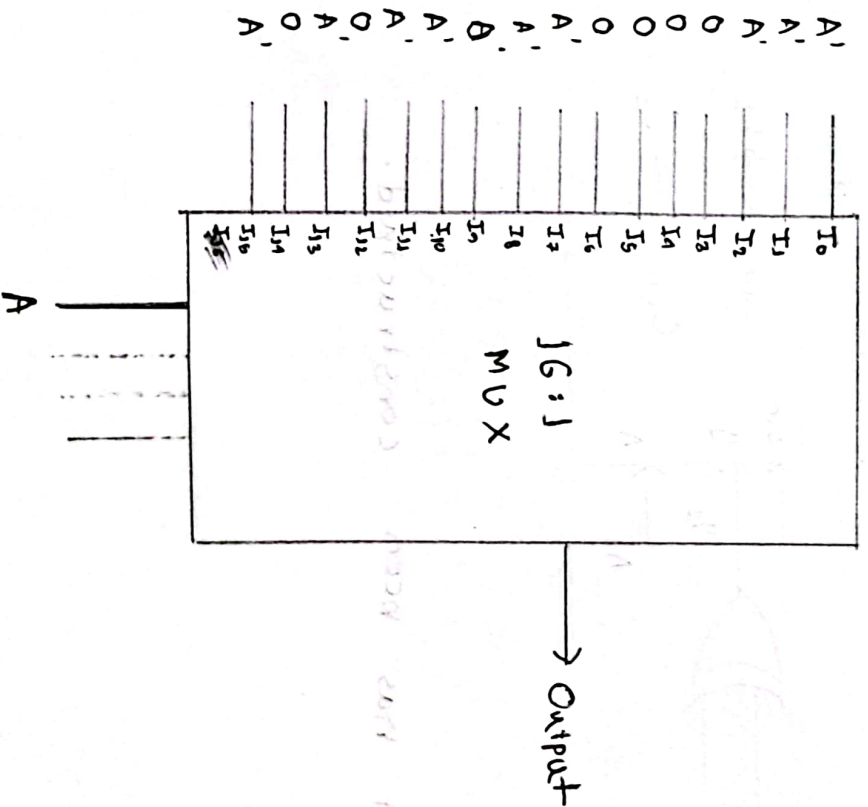
So, it has been constructed.

Answer to the Question No. 2 (a)

The function that should be implemented is $F(A, B, C, D) = \sum (0, 1, 2, 7, 8, 10, 11, 13, 15)$

Looking with (a) 16:1 mux (single)

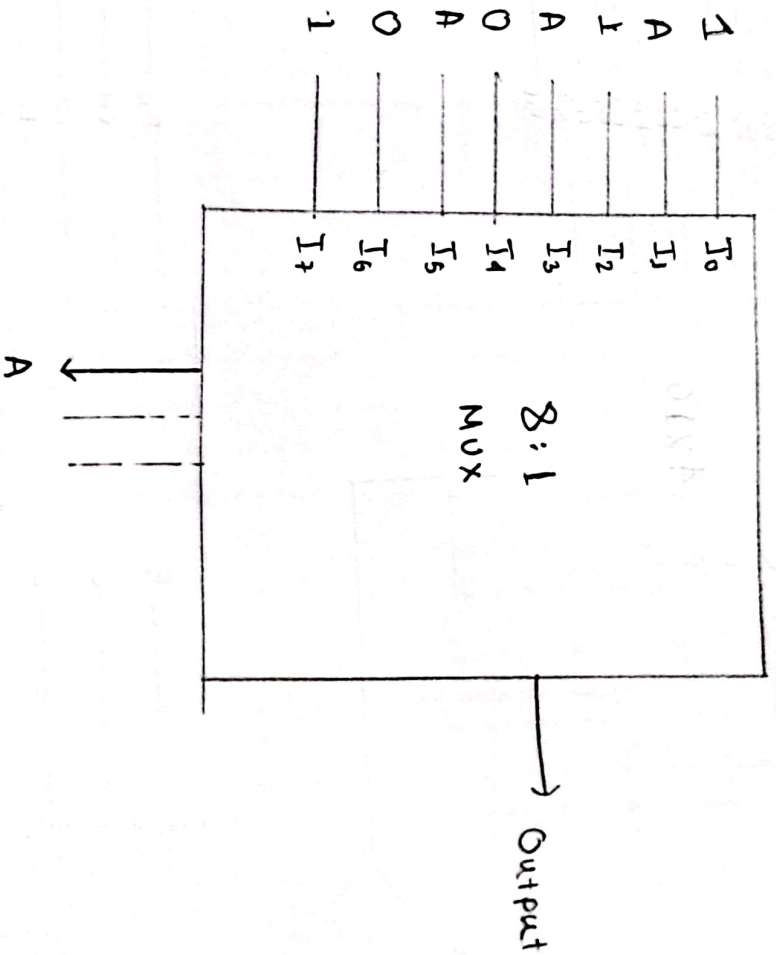
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}	I_{16}
A'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A																	
A'	A'	A'	A'	0	0	0	0	0	A'	A'	0	0	A'	0	A'	0	A'



(b)

Single 8:1 MUX

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
				3	4	5	6	7
A'	⑩	④	②					⑦
A	⑧	9	⑩	⑪	12	⑬	14	⑮
	1	A'	1	A	0	A	0	1



Answer to the Question No. 3 (a)

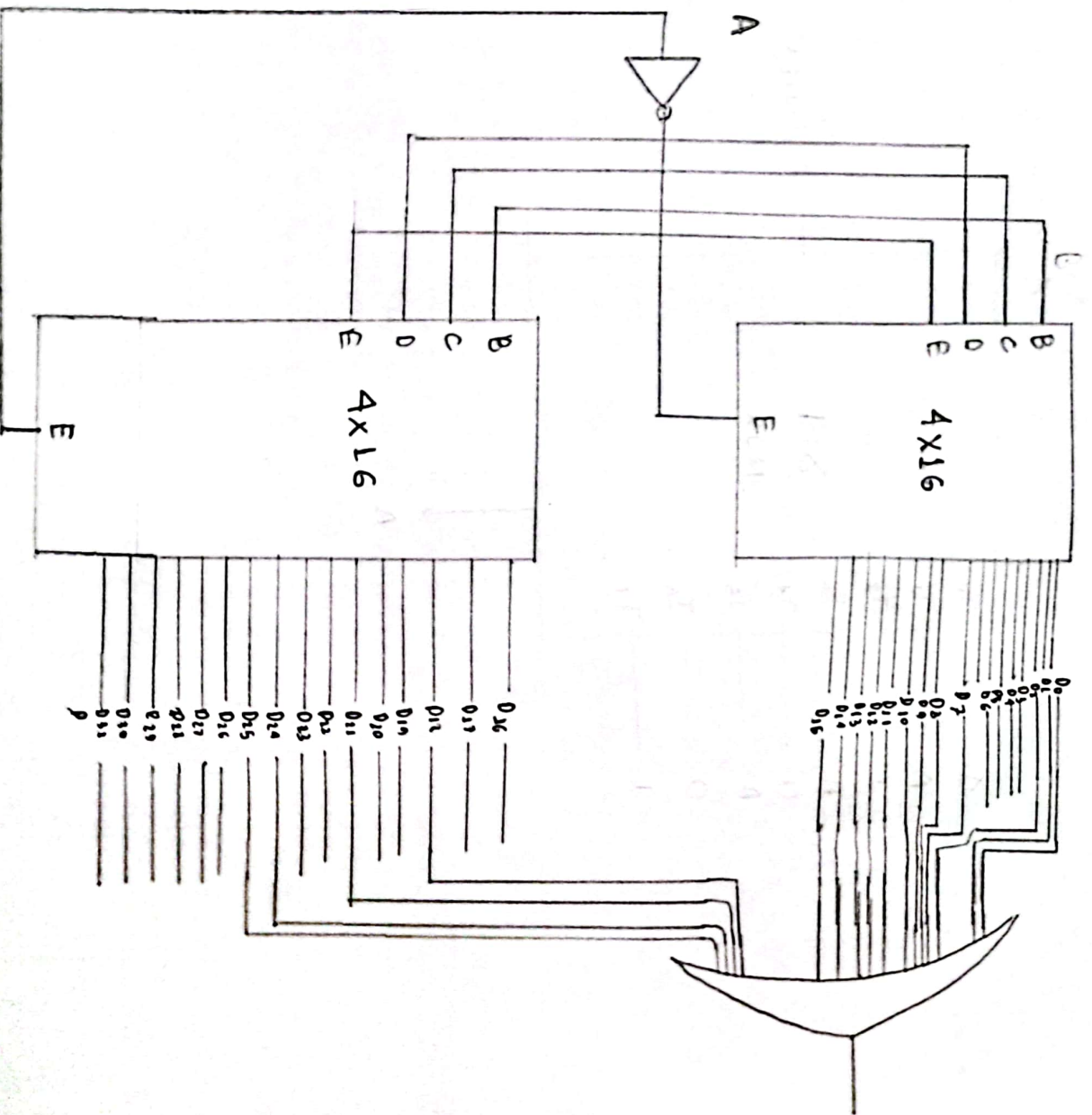
(a) Using 4 x 16 Decoders only

$$\text{The } F(A, B, C, D, E) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15, 18, 21, 24, 25)$$

We can notice there are 5 variables. So.

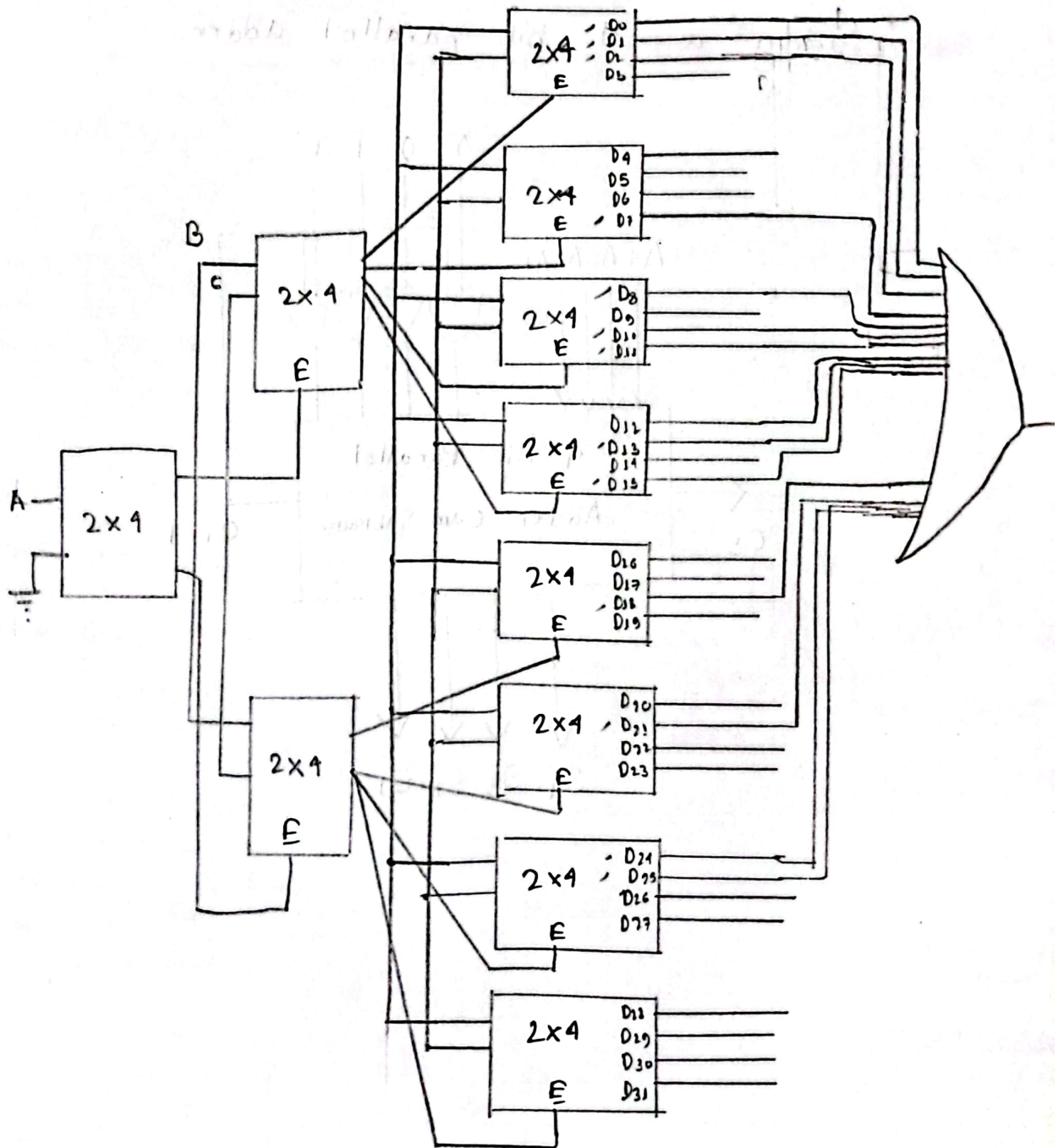
$$\text{Outputs} \rightarrow 2^5 = 32 \quad \therefore \text{So, } (16 + 16) = 32$$

Thus we need 2, 4 x 16 decoders to implement.



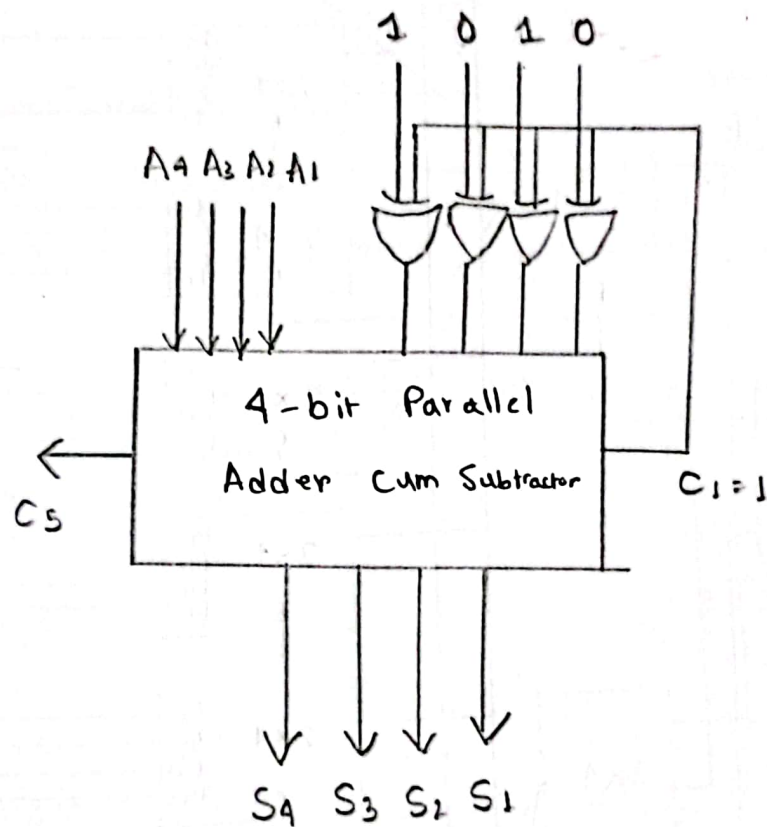
(b)

With 2x4 Decoders only



Answer to the Question No. 5

we need to implement a 4 bit number using a 4 bit parallel adder



Answer to the Question No. 4

We are required to implement a 13 person voter system using full and parallel adders

